

Assignment - Regression Algorithm

1. In this problem statement, the input and output variables are clearly defined. Therefore, this problem is solved using a Machine Learning Supervised Regression approach.
2. The dataset contains 1,338 rows and 6 input features. The input variables are Age, Sex, BMI, Children, and Smoker, while the output (target) variable is Charges.
3. During the data preprocessing stage, the categorical (string) variables were converted into binary numerical values using one-hot encoding (dummy variables), as they represent nominal data.
4. I developed a Machine Learning Supervised Regression model using the Decision Tree Regressor algorithm. The model achieved an R^2 score of 0.88, indicating good prediction performance on the test dataset.
5. The table below presents the regression algorithms implemented in this project along with their corresponding **R^2 scores** for performance comparison.

S.No	Criteria	Features	Splitter	Rscore
1	Multi Linear			0.78
2	SVM			0.05
3	Square error		Default	0.69
4	fredman_mse		Best	0.65
5	fredman_mse		Random	0.7
6	absolutr error		Best	0.65
7	fredman_mse	max_features=sqrt	Best	0.67
8	Random forest	n_estimators		0.85
9	fredman_mse	max depth	Best	0.88